Figure 2: Apache JMeter

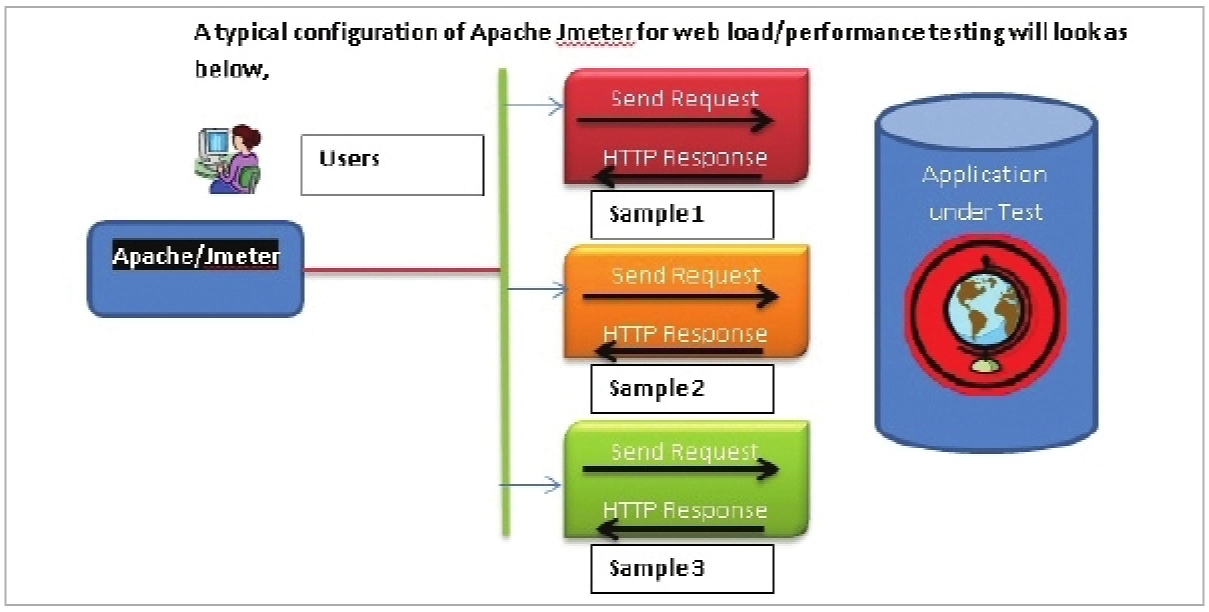

Figure 3: A typical configuration of Apache JMeter for Web load/performance testing

lot of unnecessary calls.

5) Always enabling prefetching.
6) Making sure that the CDN (content delivery network) for the Web application is configured properly and, if required, increasing the speed of a CDN and caching.
7) It is always a good practice to compress files and optimise images.

In addition to the above measures, one can use a better tool or toolset to measure Web application performance. Before finalising or shortlisting any tool, one has to consider load times, the value outcome of performances to be measured and what was the prior experience, along with metrics of the performance testing. A clear picture of these aspects is a must.

Figure 1 shows the graph depicting Web page growth over the years, of various types of content.

Some of the popular open source tools for Web optimisation are featured below.

JMeter: A load and performance tester

JMeter *(https://jmeter.apache.org/)* is a Java based open source tool supported by the Apache Software Foundation. It was developed and designed to load-test functional behaviour and measure the performance of any website or Web application. It was originally designed for testing Web applications but has since been enhanced to offer other test functions.

When we need performance across the complete Web environment, we need to have a proper configuration setup to carry out thorough evaluation. The typical configuration of Apache JMeter for Web load/performance testing will look like what's shown in Figure 3.

The key features of JMeter are:

- The ability to test the load and performance of many different applications, servers and protocols along with Web based applications.
- Supports all Web standards like HTTP, HTTPS (Java, Node.js, PHP, ASP.NET, etc)
- Supports all major APIs like SOAP/REST Web services and Java objects
- Supports File Transfer Protocol (FTP) and TCP
- Databases are supported using JDBC
- Security features like LDAP are supported
- Message-oriented middleware (MOM) via JMS
- Protocols like Mail-SMTP(S), POP3(S) and IMAP(S) are supported
- Supports both native commands or shell scripts

Figure 4: Grinder

Grinder: A Java load-testing framework

Grinder *(http://grinder.sourceforge.net/)* is a Java load testing framework. Its features make it easy to run a distributed test using many load injector machines. Grinder is free and is available under a BSD-style open source licence.

Some of the key features of Grinder are listed below:

- *Generic approach load test:* This means that anything that has a Java API will be supported, including HTTP Web servers, SOAP and REST Web services, as well as application servers (CORBA, RMI, JMS, EJBs, etc) and custom protocols.
- *Grinder supports flexible scripting:* In most cases, test scripts are written in the powerful Jython and Clojure languages. This helps in developing test cases and powerful test architecture.
- *Grinder comes with a distributed framework:* Since it has a graphical console, this allows multiple load injectors to be monitored and controlled, and provides centralised script editing and distribution.
- *Mature HTTP support:* Automatic management of client connections and cookies.
- *Dynamic scripting:* Test scripts are written using a dynamic scripting language and specify the tests to run. The default script language is Jython (which is a Java implementation of the popular Python language).
- Supports SSL, a proxy-aware